Yibo Yuan

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RESEARCH PROFILE

I study whether and how non-biological social partners - embodied or virtual - can alleviate loneliness and satisfy social need. Across hypothalamic population dynamics, programmable social stimuli, human-subjects HCI, and authority-safe AI memory, I ask which neural, sensory, and relational mechanisms make artificial companionship effective without overstepping human boundaries.

EDUCATION

Xi'an Jiaotong University, Xi'an, China **Sep 2021 - Jun 2025**
B.Eng. in Biomedical Engineering (Medical Electronic Information), School of Life Science and Technology

- First-year Chemical Biology program, Qian Xuesen Honors College (2021-2022), before transferring to Biomedical Engineering.

RESEARCH EXPERIENCE

Systems Social Neuroscience Lab, Westlake University, Hangzhou, China **Aug 2025 - Present**
Research Assistant; supervised by Prof. Ding Liu

Research program: Can non-biological social partners - embodied or virtual - alleviate loneliness and satisfy social need, and which neural, sensory, and relational mechanisms determine when they succeed?

Neural Representation and Computational Modeling of Social Need **Nov 2025 - Present**

- Study how social isolation and reunion reshape medial preoptic nucleus (MPN) population dynamics by aligning miniscope calcium imaging with frame-level social contact behavior.
- Independently developed an open-source, high-throughput semantic-segmentation pipeline for social-contact quantification across irregular arenas and heterogeneous video quality, with perspective correction, temporal alignment, and frame-level outputs.
- Developed a registration-free population-geometry framework for cross-session and cross-day comparison and found that behaviorally inferred social need covaries with MPN population sensitivity during reunion, supported by cross-validated decoding, motion controls, and circular-shift and permutation tests.

Programmable Biomimetic Robotic Mouse *- Decomposing the sensory variables of social need* **Jan 2026 - Present**

- Conceived the project and assembled an interdisciplinary collaboration to use a programmable embodied partner as a standardized social stimulus, independently manipulating motion, touch, temperature, olfactory cues, and responsiveness.
- Designed a cue-decomposition and reunion-probe paradigm plus a safety-constrained, high-throughput sound-attenuated platform integrating multi-camera perception, active/passive interaction, synchronized video, ultrasonic-vocalization recording, and wall-aware navigation.

Controlled-Touch Calibration of Social-Need Accumulation *- Experimental tool and modeling paradigm* **2026 - Present**

- Designed a controlled-touch paradigm using programmable contact number, timing, duration, and tactile properties as calibrated inputs for comparing leaky-integrator, HMM-AR, attractor, and social-avoidance models of how social need accumulates and decays.

MORI: Relational Power in Human-AI Companionship *- CHI-targeted human-subjects study* **2026 - Present**

- Conceived and led a 105-participant experiment on how people grant, revise, and withdraw distinct powers from an AI companion in long-term relationships and loneliness-support contexts, in ongoing collaboration with Xiaohongshu.
- Designed a fixed 20-year interactive family narrative spanning six powers, four governance arrangements, and three judgment stages; captured final choices, within-trial revisions, decision times, and open-ended explanations.
- Found power-specific and often bidirectional boundary revision; memory-sharing authority showed one of the largest attitude shifts and greatest boundary volatility.

TAS-Gate: Toward More Socially and Emotionally Intelligent AI *- Computational extension of MORI* **2026 - Present**

- Built on MORI's finding that AI authority is power-specific: proposed asymmetric rule and experience memories plus a structured authorization gate so long-term AI can learn from success, praise, and trust without mistaking them for permission to expand its authority.

Cross-lab Collaboration: AI for Scientific Simulation and Discovery Lab *- fMRI dynamics* **Mar 2026 - Present**

- Work with Prof. Tailin Wu and Ph.D. candidate Tengfei Xu, in collaboration with Prof. Pulin Gong (The University of Sydney), to develop models that generate cortical phase singularities and validate them with BrainVortexToolbox through phase-gradient and vortex-trajectory analyses.

Trustworthy and Intelligent Machine Learning Lab, Illinois Institute of Technology, Remote **Nov 2025 - Present**
Remote Research Collaborator; supervised by Prof. Ren Wang ***Reliable Cross-subject EEG-fNIRS Fusion***

- Developed and rigorously evaluated an EEG-fNIRS heterogeneous-expert framework for cross-subject brain-state decoding under participant-disjoint, few-shot protocols.
- Achieved 74.01% balanced accuracy on the 26-participant Shin2018-WG benchmark and used external failure analysis to identify when unreliable modalities degrade fusion, motivating reliability-aware gating and safe fallback.

SELECTED UNDERGRADUATE RESEARCH AND ENGINEERING

Portable Bioimpedance Spectroscopy System | 2025: Graduation project supervised by Profs. Xiang Chen and Jin Li; designed and bench-tested an STM32/AD5933 tissue-water-content sensing prototype based on multi-frequency bioimpedance and the Cole-Cole model.
Eight-channel EEG Acquisition System | 2024: Led PCB architecture and circuit design for an ADS1299/STM32F103 system, including power/reference, input protection, anti-aliasing, bias drive, and test-signal paths.
XJTU iGEM | 2022: Supported plasmid design, PCR/RT-PCR, nucleic-acid purification, Western blotting, electrophoresis, and literature analysis for a synthetic-biology project; team received an iGEM Gold Medal.

PUBLICATION AND RESEARCH OUTPUTS

Published

Dai, Y. Y., Wei, C., Yuan, Y., Rahman, M. M., & Liu, D. (2026). In vivo calcium imaging with a miniaturized microscope in the hypothalamus for understanding social behaviors in mice. *Journal of Visualized Experiments*, (229), e70401. doi:10.3791/70401

RESEARCH SOFTWARE AND EXPERIMENTAL PROTOTYPES

Within Prof. Ding Liu's lab, I translate diverse experimental requirements from the lab and its collaborators into customized, deployable research software, interactive tools, embedded systems, and experimental apparatus.

Mouse-trajectory-tracking **| Open source** - High-throughput semantic-segmentation workflow for social-contact quantification across irregular arenas and heterogeneous video quality.
Lickometer **| Open source** - Eight-channel STM32 capacitive lickometer with embedded firmware and a Python interface for real-time visualization and quantitative export.
NeuronTracker **| Open source** - Human-in-the-loop axon-growth tracing tool developed for a collaboration with Prof. Rui Fang's lab, using biologically motivated path constraints.
Macaque behavioral analysis | Customized prototype: Mask-based 2.5D workflow developed for a collaboration with Prof. Xiaodong Liu's team, including cage calibration, planar trajectory proxies, activity fragmentation, and spatial occupancy.
Motion-sickness experimental platform | In-lab prototype: Programmable three-axis motion platform with configurable motion profiles, multimodal synchronization, and custom 3D-printed components.

PROFESSIONAL EXPERIENCE

Zhen Tec Technology Inc. | EEG Data Analyst | Jun-Jul 2024: Designed EEG filters and time/frequency features with MNE and supported PCA/ICA and CNN-LSTM fatigue-model development.
Collegeland Education Co. Ltd. | AI Assistant Developer | May-Sep 2024: Built data-processing tools and supported prompting, hallucination handling, and adversarial testing for educational LLM workflows.
First Affiliated Hospital of Xi'an Jiaotong University | Clinical Laboratory Intern | May-Jun 2023: Observed rehabilitation and imaging workflows and evaluated usability constraints in clinical equipment.

HONORS AND AWARDS

2024: National College Students' Innovation and Entrepreneurship Training Program; Second Prize, 35th Tengfei Cup Innovation Track.
2023: National College Students' Innovation and Entrepreneurship Training Program; Gold Prize, 34th Tengfei Cup Entrepreneurship Track; Second Prize, XJTU Mathematical Modeling Competition.
2022: Gold Medal, International Genetically Engineered Machine Competition (iGEM).

LEADERSHIP AND SERVICE

Medical and Engineering Innovators | Council Member, 2024-2025: Led alumni liaison, supported a Technology and Finance forum, and proposed a hospital-university translational project.
The Micro Light Tutoring Association | Volunteer Lecturer, 2023: Provided free one-on-one online tutoring to high-school students in Tibet.

TECHNICAL SKILLS

Programming and ML: Python, C, MATLAB, PyTorch, scikit-learn, pandas, NumPy, MNE, OpenCV, MMSegmentation, OpenMMLab, CEBRA, React, TypeScript
Neural and behavioral analysis: Miniscope calcium imaging, neural manifolds, population decoding, representational similarity, hazard analysis, long-duration tracking, semantic segmentation, SAM2, SLEAP
Neuroengineering and hardware: STM32, ADS1299, AD5933, UART, PCB design, soldering/debugging, stepper-motor control, Blender, 3D printing, experimental synchronization
Wet laboratory: Mouse stereotactic surgery, injection, perfusion, brain cryosectioning, optogenetic preparation, DNA/RNA extraction, PCR/RT-PCR, Western blotting
Research tools: Git/GitHub, Linux, Cloudflare, Bonsai, Miniscope GUI, SnapGene, Adobe Illustrator, SPSS
Languages: Mandarin Chinese (native); English (professional working proficiency)